Reduction of Order (Week 6)

Spiegel "Applied Differential Equations", P. 153

Find the general solution of

$$\frac{d^2y}{dt^2} - t\frac{dy}{dt} + y = 0 \qquad (1)$$

given that $y = t$ is a solution.

Solution Set

$$y = tv \qquad (2)$$

In (1),

$$\frac{dy}{dt} = \frac{d}{dt}(tv) = v + t\frac{dv}{dt} \qquad (3)$$

$$\frac{d^2y}{dt^2} = \frac{dv}{dt} + t\frac{d^2v}{dt^2} + \frac{dv}{dt}$$

$$= 2\frac{dv}{dt} + t\frac{d^2v}{dt^2} \qquad (4)$$

Substituting (2), (3) into (1) yields

$$t\frac{d^2v}{dt^2} + 2\frac{dv}{dt} - tv - t^2\frac{dv}{dt} + tv = 0$$

$$t\frac{d^2v}{dt^2} + (2 - t^2)\frac{dv}{dt} = 0 \qquad (5)$$

Eq. (1) is a linear ODE of 2nd order. Eq (5) is a linear ODE of 1st order in $\frac{dv}{dt}$, so we reduced the order.

Setting

$$z = \frac{dv}{dt} \qquad (6)$$

in (5) yields

$$t\frac{dz}{dt} + (2-t^2)z = 0$$

which is clearly first-order

$$\frac{1}{z}dz + \frac{2-t^2}{t}dt = 0 \qquad t \neq 0$$

$$\int \frac{1}{z}dz + \int\left(\frac{2}{t} - t\right)dt = C_1$$

$$\ln|z| + 2\ln|t| - \frac{t^2}{2} = C$$

$$|z|\,t^2 e^{-t^2/2} = C$$

$$z = C_1 e^{t^2/2}/t^2$$

From (6),

$$v = \int z(t)\,dt = C_1\int e^{t^2/2}\left(\frac{1}{t^2}\right)dt + C_2$$

Hence $y(t) = tv = C_2 t + C_1 t\left[\int e^{t^2/2}\left(\frac{1}{t^2}\right)dt\right]$

Method of Undetermined Coefficients

M. R. Spiegel "Applied Differential Equations" Pg 158

Purpose:
To obtain a particular Solution.

Solve $y'' + y = 6\cos^2 t$, (1)
given $y(0) = 0$, $y(\pi/2) = 0$

1. Find complementary solution, y_c the general solution to

$$\ddot{y} + y = 0 \tag{2}$$

$$y_c = C_1 \sin(t) + C_2 \cos(t). \tag{3}$$

$$y(t) = y_c(t) + y_p(t) \tag{4}$$

Recall

$$\cos^2(t) = \frac{1}{2} + \frac{1}{2}\cos(2t)$$

so

$$6\cos^2(t) = 3 + 3\cos(2t) \tag{5}$$

Assume

$$y_p(t) = A + B\cos(2t) + C\sin(2t) \tag{6}$$

where coefficients A, B, C are to be determined.

$$y_p'(t) = -2B\sin(2t) + 2C\cos(2t) \tag{7}$$

$$y_p''(t) = -4B\cos(2t) - 4C\sin(2t) \tag{8}$$

Substituting (6), (8) into (1) yields

$$A + B\cos(2t) + C\sin(2t) - 4B\cos(2t) - 4C\sin(2t)$$

$$= 3 + 3\cos(2t) \qquad (9)$$

if we take (5) into account

Equating the constants and the corresponding coefficients on the left and right sides of (9) yields

$$A = 3$$

$$-3B = 3 \implies B = -1$$

$$-3C = 0 \implies C = 0$$

From (6)

$$y_p(t) = 3 - \cos(2t) \qquad (10)$$

From (4)

$$y(t) = 3 - \cos(2t) + C_1\sin(t) + C_2\cos(t)$$

$$y(0) = 0 \implies 3 - \cos(0) + C_2\cos(0) = 0$$

$$2 + C_2 = 0 \implies C_2 = -2$$

$$y(\pi/2) = 0 \implies 3 - \cos(\pi) + C_1\sin(\pi/2) + C_2\cos(\pi/2) = 0$$

$$4 + C_1 = 0 \implies C_1 = -4$$

So

$$y(t) = 3 - \cos(2t) - 4\sin(t) - 2\cos(t)$$